Lecture 8

Thursday, 2 February 2023 1:30 PM

What is α, the transfer coeff.?

For a multi-step/multi-e⁻ electrode rxn, $A + ne^- \rightarrow Z$, composed of n e⁻ transfer steps where $O + re^- \rightleftharpoons R$ is RLS (stoichiometric coeff.)

w/ stoichiometry $Z = \gamma P$, and assume s e⁻ transferred before RLS and s' e⁻ are transferred after RLS

$r = 0 \rightarrow$ chemical step
$r = 1 \rightarrow$ e-chem step
} No. of e⁻ in rate limiting step

$|i_c| = F \cdot c_0 \cdot \exp\left[\frac{-r\beta\eta F}{RT}\right]$, and after applying quasi-equ. for all other steps:

$$i = i_0 \left[\exp\left(\left(\frac{n-s}{\gamma} - r\beta\right)\frac{F\eta}{RT}\right) - \exp\left(-\left(\frac{s}{\gamma} + r\beta\right)\frac{F\eta}{RT}\right)\right] \quad ; \quad \alpha_a = \frac{n-s}{\nu} - r\beta$$

$n\nu = s + s' + r$

Independent rxn:

$$\alpha_c = \frac{s}{\gamma} + r\beta$$

If RLS is non-faradaic, $r = 0 \rightarrow \beta$ does not appear in α_a, α_c but will depend on potential

$$\alpha_a + \alpha_c = \frac{n}{\nu}$$

Kinetics of Multiple Rxns

↳ More than 1 rxn can occur @ electrode simultaneously ; ex: CO_2 electrochemical reduction to CO

$CO_2 + H_2O + 2e^- \rightarrow CO + 2OH^-$ ← This reaction occurs in the same potential regime as water electrolysis $[2H_2O + 2e^- \rightarrow H_2 + 2OH^-]$

† The total current observed would be sum of partial currents:

$i_T = i_{CO} + i_{H_2}$ → we can model each partial current w/ B-V expression, assuming they are non-interacting

↳ We might also assign a "current efficiency" or Faradaic efficiency to understand selectivity

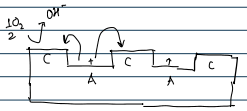
Is every process Faradaic?
↓
How can non-faradaic process occur
↓
VdW? But then that is not a bond?
e⁻ coming from the same element)

Corrosion → special type of multiple rxn

↳ What must net current of corroding metal be?

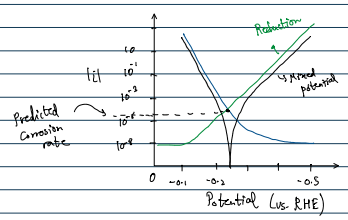
$I_{total} = 0$, $i_{ox} = i_{red}$; other reactions: $H_2O + 2e^- \rightarrow H_2 + 2OH^-$

$H_2O + \frac{1}{2}O_2 + 2e^- \rightarrow 2OH^-$



We care about the potential at which this process happens:
→ We polarize our electrode to find this point

$Fe \rightarrow Fe^{2+} + 2e^-$
$2H_2O + 2e^- \rightarrow 2OH^- + H_2$



→ Reducing potential → helping the reduction reaction

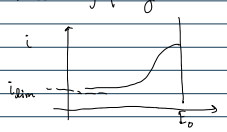
→ For non-zero currents → you need counter electrode??

Alloy electroplating

↳ We desire multiple rxns

Non-idealities that deviate from B-V

1) Mass Transport (ch. 5): We can have large changes in i w/ small change in V so we frequently move b/w non-limited + MT limited regions



2) Passivation: Certain oxidized metals are not soluble, & therefore they do not react further

ex: $Cu \rightarrow Cu^{2+} + 2e^-$

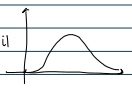


Stability of electrodes this

2) Passivation : Certain oxidized metals are not soluble, & therefore they do not react further

ex: $Cu \rightarrow Cu^{2+} + 2e^-$

$Cu \rightarrow CuOH_{(s)} + e^-$



Stability of electrodes is determined from this

We can understand regimes w/ Potential - pH or Pourbaix diagram

3) Organic additives : Often used in electroplating to achieve desirable feature filling

Not taking part in e-chem reaction but are in fact affecting the rate (need to incorporate these non-reacting species into C_i ~~reaction~~)

4) Electrode poisoning : Chem species bind so strongly that they eat up active sites + don't leave $CO + Pt \rightarrow Pt-CO$